

# Urba Zafar

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## HONORS & AWARDS

- **Chancellor's Bronze Medal** | Awarded for outstanding academic performance in undergraduate.
- **Dean's List of Honors Award** | Secured a perfect **4.00/4.00 SGPA** in the 8th semester.
- **Ranked 3rd among 90 students** | Maintained an overall **CGPA of 3.63/4.00** in BS Bioinformatics.
- **IELTS Academic** | Achieved 7.5 Bands in IELTS, indicating CEFR C1 Proficiency Level [[Certificate](#)]

## EDUCATION

**Government College University Faisalabad (GCUF) | HEC RECOGNIZED**

Faisalabad, PK

*Bachelor of Science in Bioinformatics*

Nov 2021 – June 2025

- **CGPA:** 3.63/4.0 | **SGPA:** 4.00/4.00 | **Rank:** 3/90

## RESEARCH

**Integrative Omics and Molecular Modeling Lab** | *Research Assistant*

August 2025 – Present

*Department of Bioinformatics & Biotechnology, GCUF*

**Principal Investigator:** [Dr. Muhammad Tahir ul Qamar](#)

- Developing a biological database integrating copy number variations (CNVs), epigenetic markers, and 3D structural data for neurodegenerative disease.
- Designing schema and pipelines to enable efficient integration, retrieval, and analysis of multi-omics data.
- Aiming to provide a comprehensive resource to support biomarker discovery and therapeutic research in neurological disease.

**Biocatalysis and Protein Engineering Research Group Lab** | *Research Assistant*

Feb 2025 – July 2025

*Department of Bioinformatics & Biotechnology, GCUF* [[Methodology](#) | [Research Report](#)]

**Principal Investigator:** [Dr. Muhammad Rizwan Javed](#)

- Conducted integrative analysis of differentially expressed genes (DEGs) and copy number variations (CNVs) in Alzheimer's and Parkinson's disease.
- Constructed protein-protein interaction networks, performed GO enrichment and pathway analysis (DAVID, KEGG) to uncover disease-relevant pathways.
- Proposed potential biomarkers and therapeutic targets to advance understanding of neurodegenerative diseases.

**Department of Computer Science** | *Research Assistant*

November 2025 – Present

*University of Idaho*

**Principal Investigator:** [Dr. Hasan Jamil](#)

- Developed an explainable Graph Neural Network (GNN) framework to identify Alzheimer's disease associated gene modules from protein-protein interaction (PPI) networks.
- Implemented GraphSAGE and GCN models with perturbation-based explainability to rank and interpret key disease genes.
- Validated top-ranked genes through KEGG pathway enrichment, Gene Ontology (GO) analysis, and GWAS overlap, confirming biological relevance.
- Achieved predictive performance of 89.6% accuracy using network topology features alone (GraphSAGE).

**Fratta Lab** | *Research Intern*

July 2026 – Oct 2026

*UCL Queen Square Institute of Neurology*

**Supervisor:** [Dr. Pietro Fratta](#)

- Investigating cryptic circRNA formation resulting from TDP-43 dysfunction in Inclusion Body Myositis (IBM), a muscle disease, using RNA-seq from IBM and control muscle biopsies.
- Detecting circRNAs via back-splice junction identification and identifying candidates significantly enriched in IBM versus control samples.

- Cross-referencing IBM-enriched circRNAs against known TDP-43-dependent cryptic splicing events from ALS/FTD datasets to assess mechanistic overlap.
- Classifying candidates as CNS-shared or muscle-specific to prioritize targets for experimental validation as stable biomarkers of TDP-43 loss of function.

SELECTED PROJECTS

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**Engineering a multi-epitope vaccine candidate against Leishmaniasis using comprehensive Immunoinformatics methods** [\[Research Report\]](#) Sep 2023 – Feb 2024

**Tools:** NCBI, VaxiJen v2.0, AllerTOP v2.0, ExPASy ProtParam, PSIPRED, SCRATCH Protein Predictor, UCSF Chimera, GRAMM-X, PDB, PDBsum, C-ImmSim.

- Designed a multi-epitope vaccine targeting *Leishmania* proteins (Gp63, Histone H1) using immunoinformatics approaches.
- Predicted HTL, CTL, and B-cell epitopes; evaluated antigenicity, allergenicity, and toxicity.
- Modeled the 3D structure, performed TLR2 docking, and validated stability through molecular dynamics simulations.

**Absolute Brainologists** Mar 2024 – Jun 2024

- Developed a responsive quiz website using HTML, CSS, and JavaScript with real-time scoring features.
- Implemented dynamic navigation across quiz categories (e.g., Science, Nature) and automated results display.
- Designed a clean, mobile-friendly UI with customizable quiz options for enhanced user experience.

TECHNICAL SKILLS

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- Programming Languages:** Python, R, C/C++, SQL, JavaScript, HTML, CSS.  
**Frameworks & Libraries:** Pandas, NumPy, matplotlib, Biopython, FastAPI.  
**Softwares & Platforms:** Linux, Jupyter Notebook, TBtools, Adobe Illustrator, Photoshop, Visual Studio.  
**Scientific Skills:** Scientific Writing, Data Visualization, Research Methodology, Statistical Analysis, Literature Review.  
**Machine Learning:** XGBoost, Random Forest, LASSO Regression, Deep Neural Networks, SHAP  
**Bioinformatics Tools:**
- **Genomics & Gene Analysis:** Galaxy, TBtools, Genomics, MEGA, SAMtools.
  - **Protein Modeling & Structure:** PyMOL UCSF, Chimera.
  - **Sequence Alignment & Phylogenetics:** BLAST, Clustal Omega, MUSCLE, BioEdit.
  - **Network & Pathway Analysis:** Cytoscape, STRING, KEGG Mapper, Reactome.
  - **Databases:** NCBI, UniProt, PDB, Ensembl, ENA, GEO, ArrayExpress, ClinVar, Answer ALS.

ACADEMIC CERTIFICATIONS

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|------------------------------------------------------------------------------------------|------------------------------------------|-----------|
| - <a href="#">Fundamentals of Data Science in Precision Medicine and Cloud Computing</a> | Stanford University                      | Dec 2024  |
| - <a href="#">Programming Basics</a>                                                     | Stanford University                      | Aug 2024  |
| - <a href="#">Cloud Computing</a>                                                        | Stanford University                      | Sep 2024  |
| - <a href="#">Research Ethics</a>                                                        | Stanford University                      | July 2024 |
| - <a href="#">Oxford–Nanopore Third Generation Sequencing Workshop</a>                   | Government College University Faisalabad | Oct 2024  |

VOLUNTEER & LEADERSHIP

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**Writer’s Club** | [Media Secretary](#) Nov 2023 – July 2025

- Managed social media presence and assisted in organizing literary and cultural events.
- Helped organize and promote literary and cultural activities on campus.
- Coordinated basic media coverage for events and wrote short posts or caption.

**Amal Academy** | [Career-Prep Fellow](#) Dec 2024 – Mar 2025

- Completed training in professional communication, leadership, and career development.
- Managed production, finances, and distribution, supporting five local schools.
- Led sustainability initiatives reaching 100+ students through eco-awareness drives.